Global Food Prices, Inflation & Nutrition

How Structural Vulnerability Shapes Who a Food Price Shock Actually Hurts

A 30-Country Analysis, 2015–2024

Executive Summary

Global food prices spiked sharply between 2020 and 2022, driven first by COVID-19 supply chain disruptions and then by the Russia–Ukraine war's disruption of grain and fertilizer exports. Prices have since eased but remain above pre-2020 levels. The critical question for policymakers is not whether this shock happened, but **why it left some countries largely unscathed while others slid into acute hunger crises**.

Across a 30-country panel, income level alone explains a meaningful share of baseline nutrition outcomes: GDP per capita is strongly correlated with lower stunting, undernourishment, and hunger severity. But income is a poor predictor of **who got hurt most by this specific shock**. When we isolate the change in hunger outcomes during the 2022–23 crisis window, neither food import dependency nor income level shows more than a weak relationship to the size of the deterioration. The countries that fared worst — Sudan, Afghanistan, Haiti — share something the trade and income data don't capture directly: **active conflict or acute state fragility**. Countries with high import dependency but functioning governance and markets weathered the shock far better than their trade exposure alone would predict.

Bottom line: conflict and governance capacity, not income or import dependency in isolation, are the dominant drivers of vulnerability to global food shocks in this dataset. Aid and resilience strategies built primarily around income or trade-exposure criteria will misallocate resources unless conflict and fragility are built in as a primary — not secondary — targeting criterion.

1. Objective & Data

This analysis examines how the COVID-19 and Russia–Ukraine war food-price shocks affected 30 countries differently between 2015 and 2024, and tests whether that variation is best explained by income level, food-import dependency, exchange-rate exposure, or conflict/state fragility.

Datasets

- Country-year panel (30 countries, 2015–2024): inflation, hunger and nutrition indicators, GDP per capita, food import dependency
- Global FAO Food Price Index (annual)
- Documented global food-crisis events with start/end years
- Regional summary aggregates

2. Methodology

The workflow follows a standard pipeline: data inspection and cleaning, feature engineering (crisis-year tagging), exploratory analysis by income group and region, correlation analysis, and targeted hypothesis testing on the shock's differential impact. Where a relationship is central to the report's conclusions, it is tested statistically (correlation coefficient and significance) rather than asserted from visual inspection alone; where a sample is too small for a confident causal claim, that is stated directly rather than implied.

3. Findings

3.1 The Global Shock: Timing and Speed

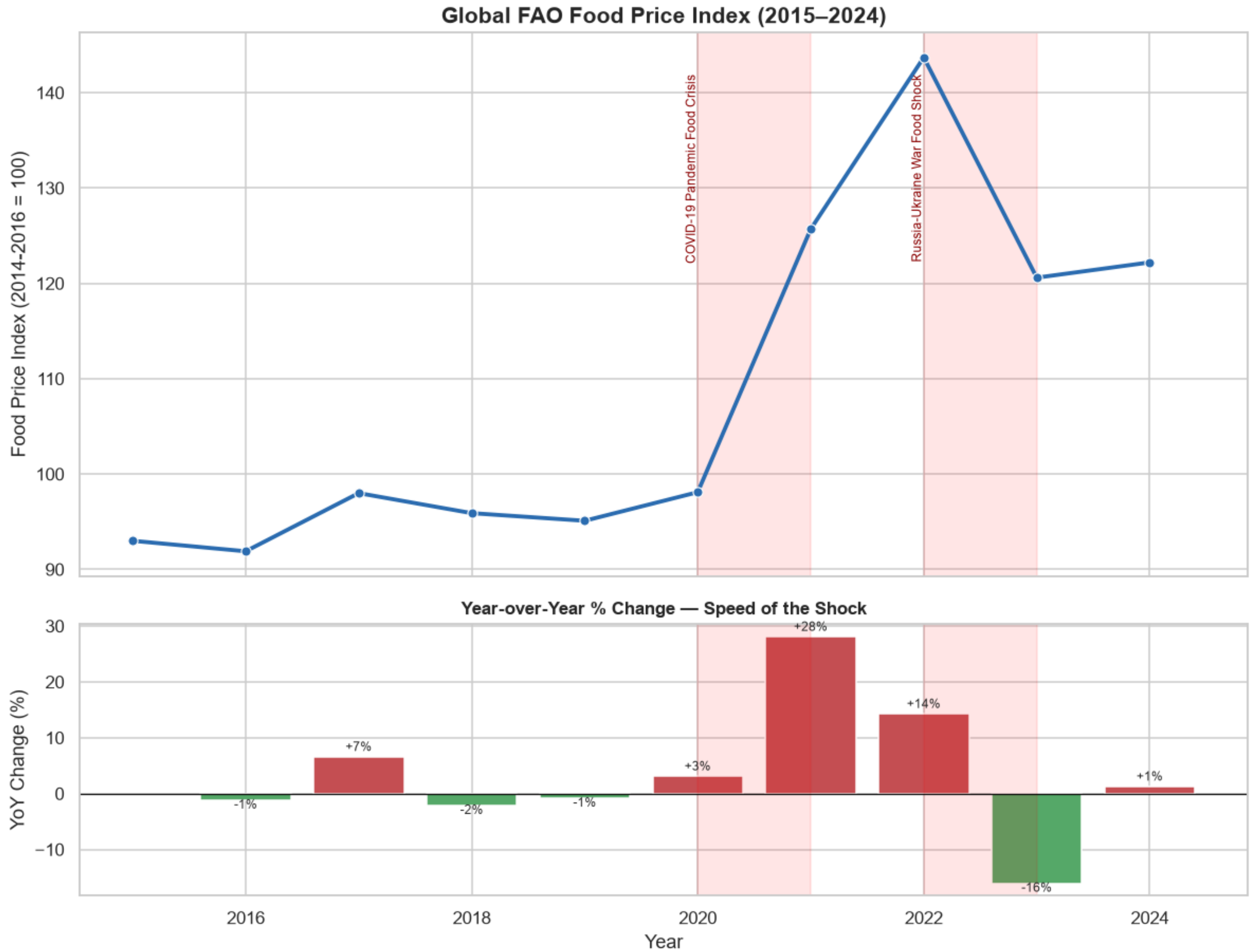

Figure 1. Global FAO Food Price Index level (top) and year-over-year % change (bottom), 2015–2024.

The index rose sharply from 2020, accelerating further through 2022 in close alignment with the Russia–Ukraine war's disruption of two of the world's largest grain and fertilizer exporters. The lower panel isolates the **speed** of the shock: 2021–2022 saw the steepest year-over-year acceleration in the ten-year window, followed by a deceleration — not a reversal — through 2023–2024. Prices remain above pre-2020 levels through the end of the observed period.

3.2 Food Inflation and Child Stunting: A Weaker Link Than It Appears

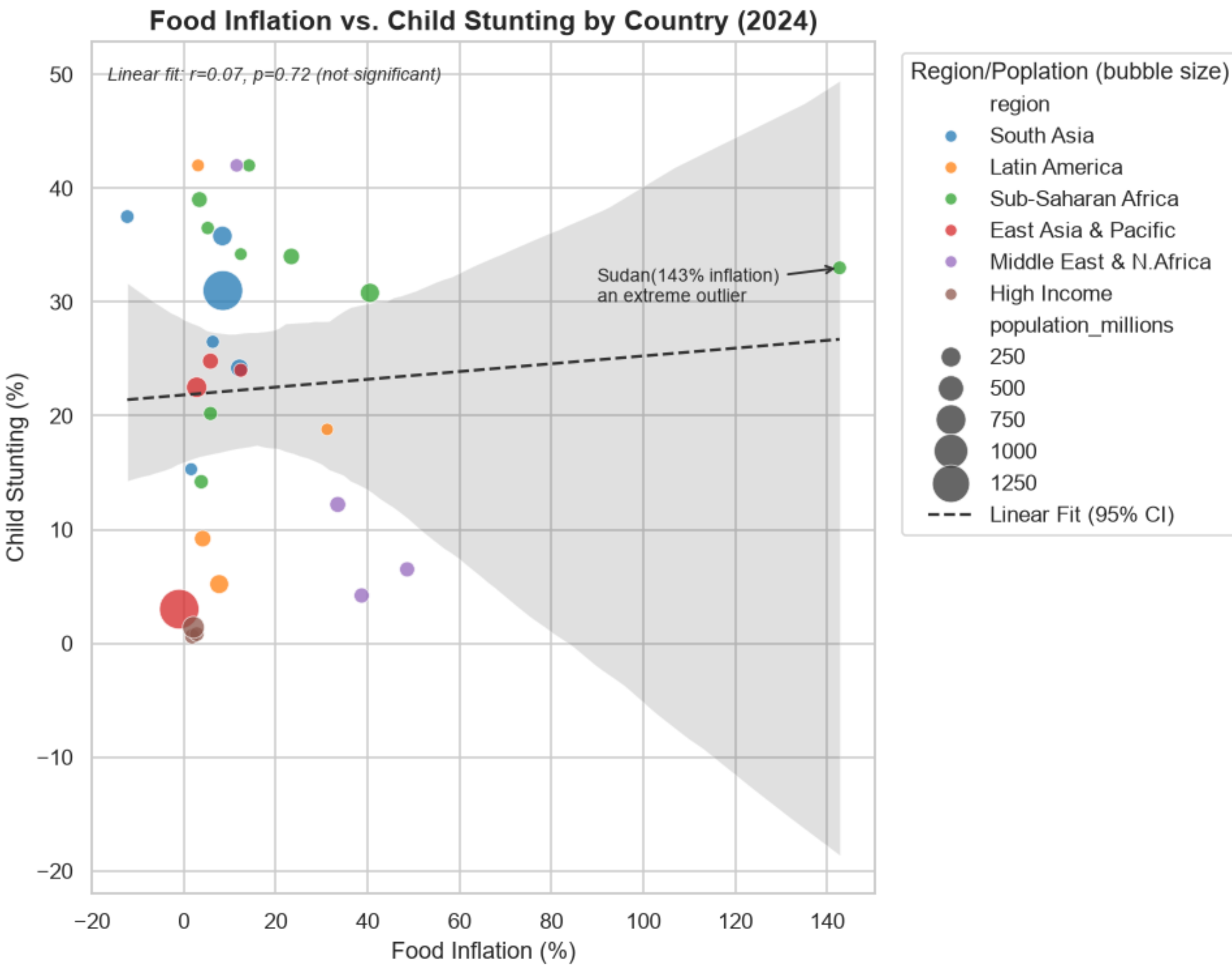

Figure 2. Food inflation vs. child stunting by country (2024), with fitted linear relationship and 95% confidence band.

At first glance, Sub-Saharan African and South Asian countries cluster toward higher food inflation and higher stunting, while high-income countries sit near the origin. But the fitted relationship across all 30 countries is **not statistically significant** ($r = 0.06$, $p = 0.75$) — food inflation in a single year does not, by itself, linearly predict child stunting in this sample. The apparent visual clustering reflects regional and income overlap rather than a direct inflation-to-stunting effect: countries with structurally high stunting also tend to have chronically volatile food prices, but the *current year's* inflation rate is not the operative variable. This distinction matters for policy — it argues against treating a single bad inflation year as a leading indicator of stunting risk, and for treating stunting as a function of structural, multi-year deprivation instead.

3.3 Regional Hunger Burden

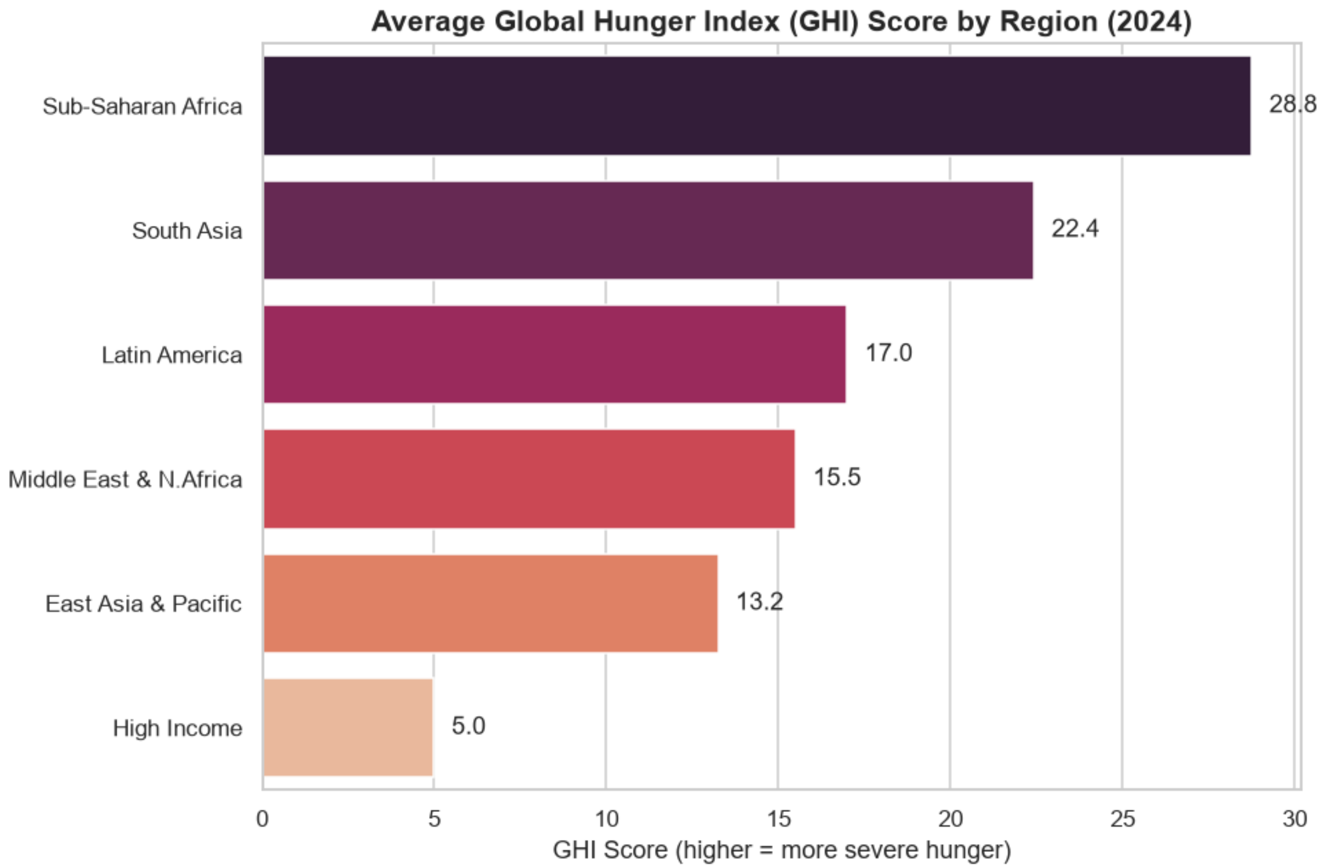

Figure 3. Average Global Hunger Index (GHI) score by region, 2024.

Sub-Saharan Africa and South Asia carry the heaviest average hunger burden. Notably, the Middle East & North Africa region shows a moderate GHI score despite the **highest average food inflation of any region (33.1%)** — a first indication that inflation exposure and hunger outcome are not tightly coupled at the regional level either, consistent with the finding in Section 3.2.

3.4 What Actually Moves Together

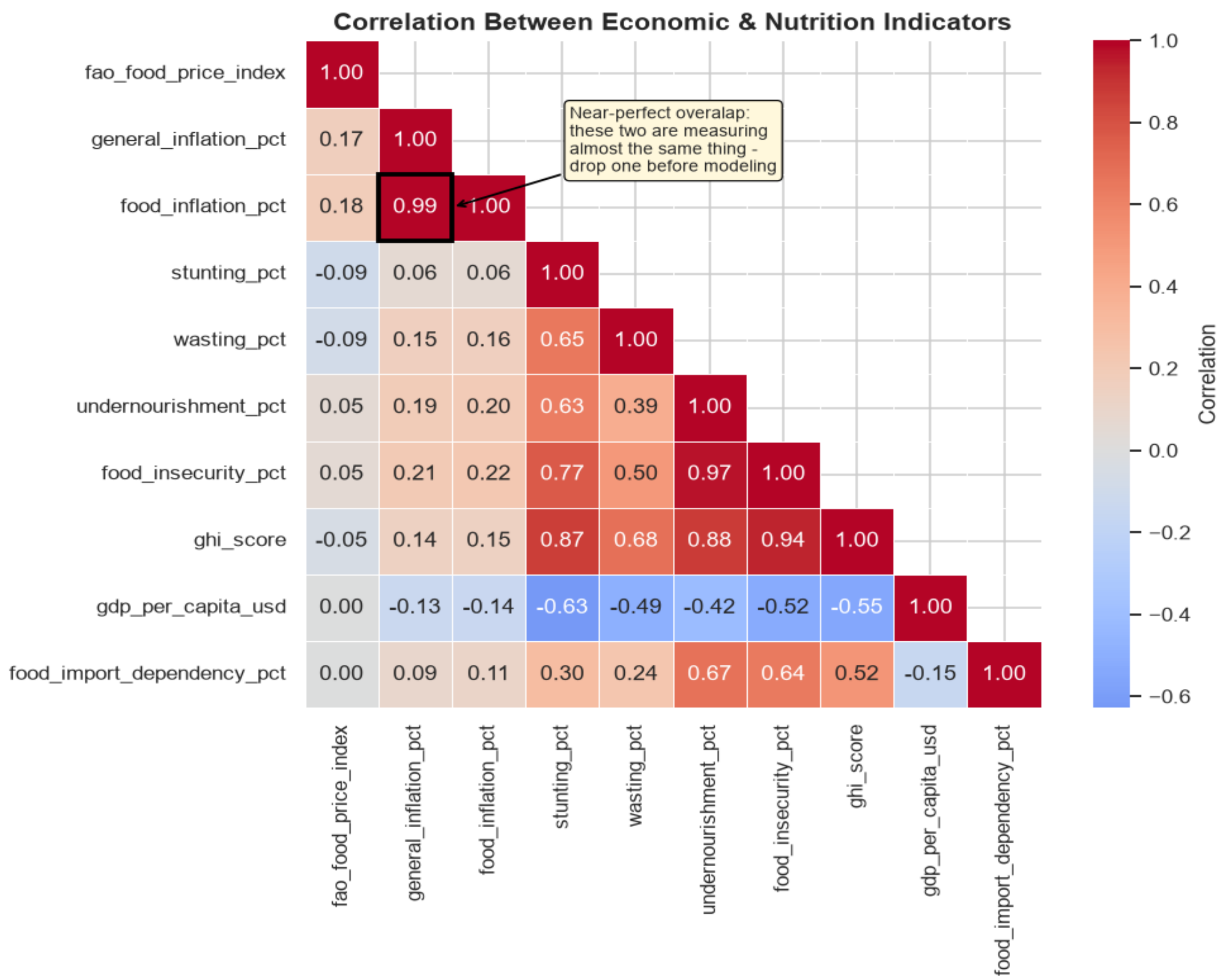

Figure 4. Correlation matrix of economic and nutrition indicators (Pearson correlation, 2015–2024 panel).

General inflation and food inflation correlate at r = 0.99 — effectively the same signal, since food inflation is a dominant component of headline inflation across this sample; the two should be treated as one variable, not two independent confirmations, in any further modeling. GDP per capita is strongly and negatively associated with stunting, undernourishment, and GHI (r = -0.55 to -0.63), and food import dependency is positively associated with undernourishment and food insecurity (r = 0.64–0.67). These are cross-sectional associations across a 30-country panel and are consistent with — but do not on their own prove — income and trade exposure as causal drivers of baseline hunger severity.

3.5 Where Food Insecurity Is Most Acute Today

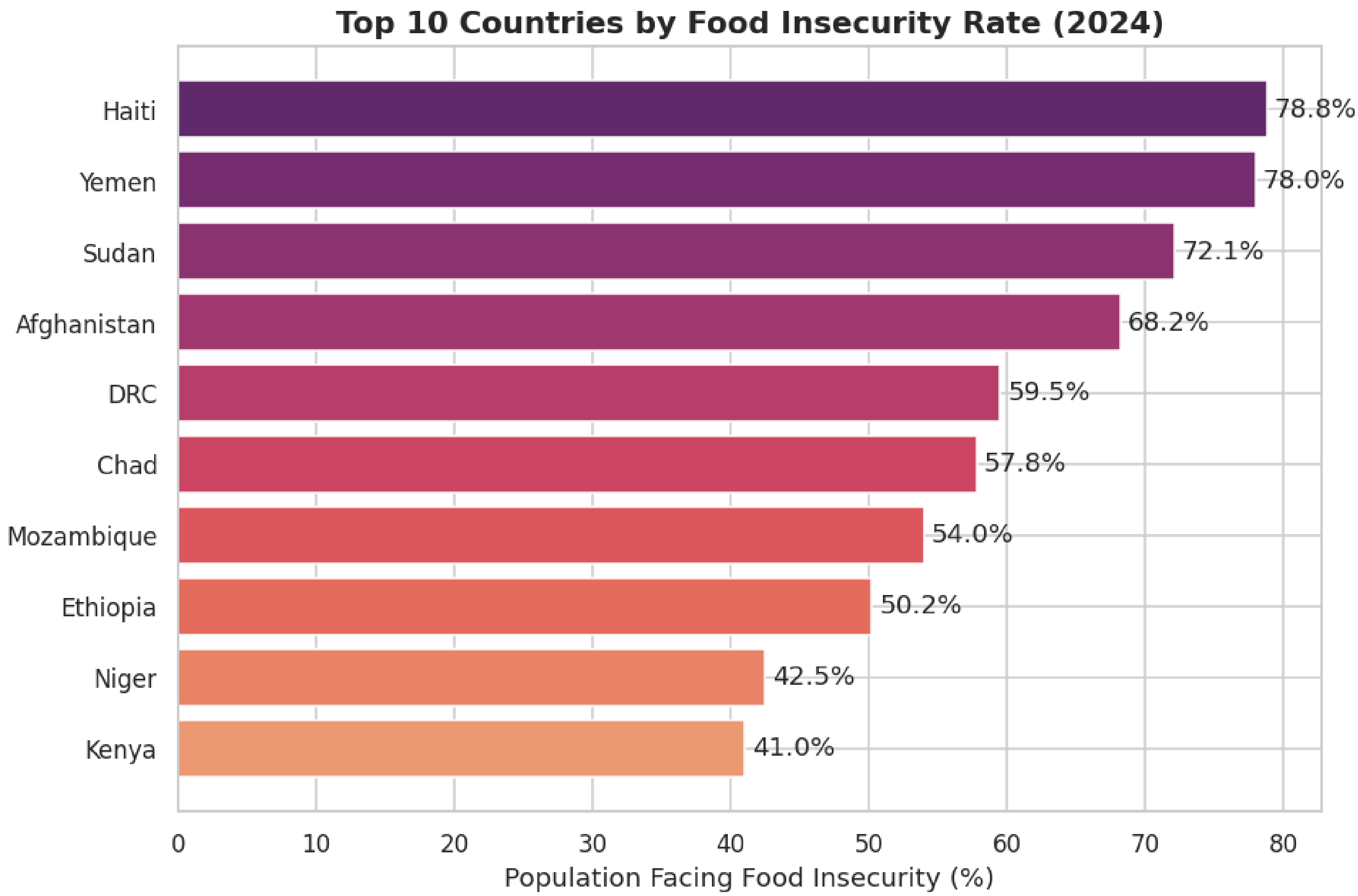

Figure 5. Top 10 countries by food insecurity rate, 2024.

These are the specific countries where the largest share of the population struggles to reliably access food today, and the natural starting point for any targeted intervention list.

3.6 Within-Group Variation Is as Important as the Average

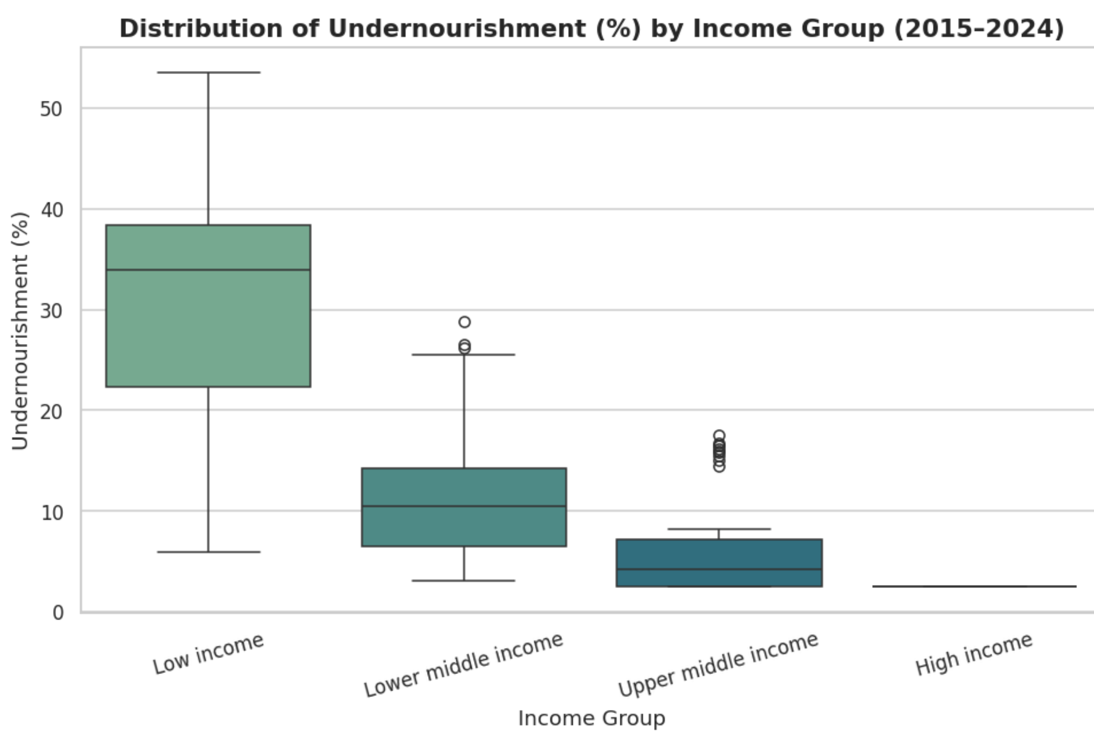

Figure 6. Distribution of undernourishment (%) by income group, 2015–2024.

Low-income countries have both the highest median undernourishment rate and by far the widest spread — some low-income countries substantially outperform their income peers, and others fall far below. Income group alone is therefore a **necessary but not sufficient** predictor; a meaningful share of the variation within each income tier is driven by something other than income.

3.7 The Real Test: What Predicts Vulnerability to a Shock?

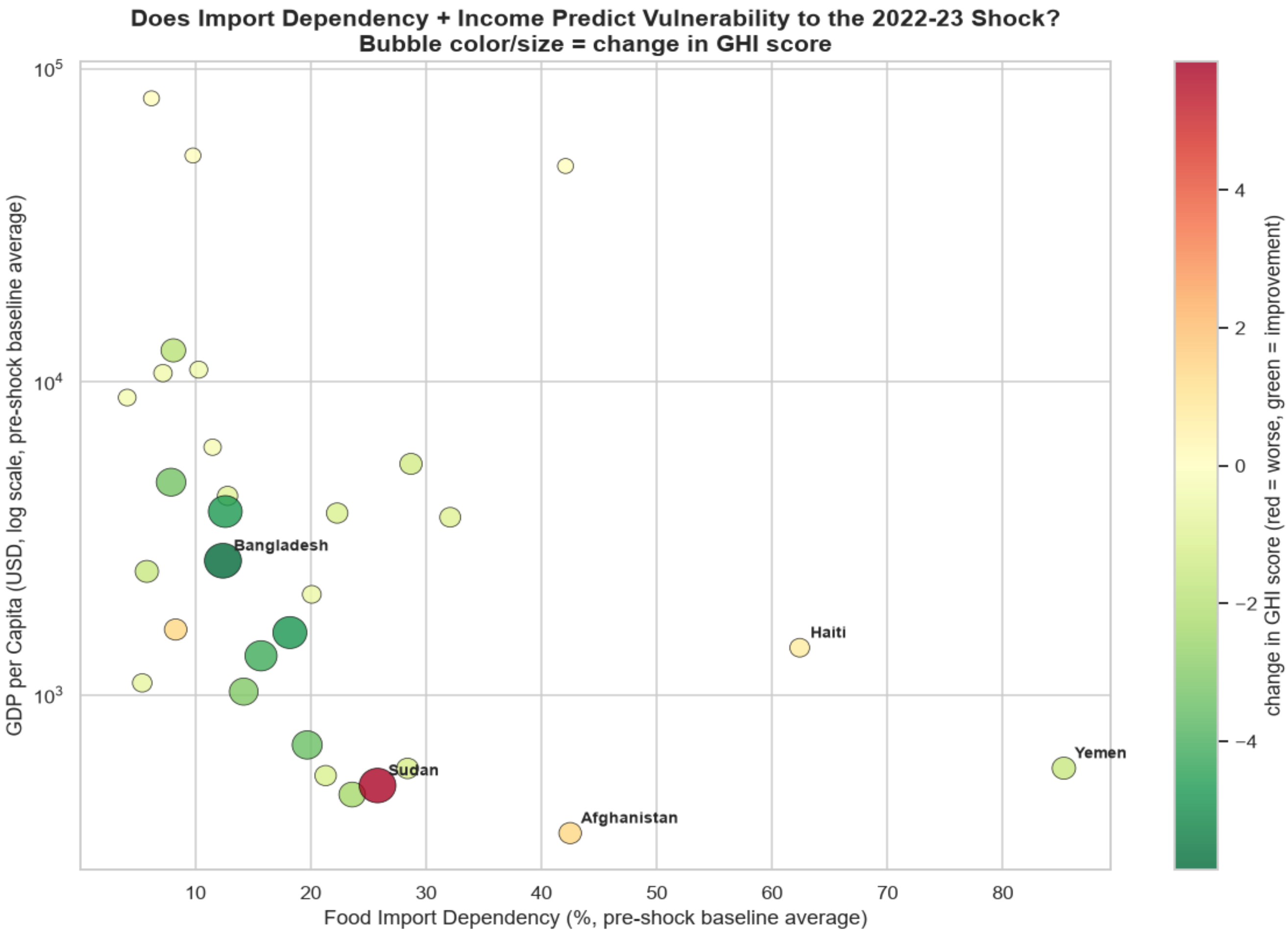

Figure 7. Pre-shock food import dependency and GDP per capita vs. change in GHI score during the 2022–23 shock. Color and bubble size encode the magnitude of deterioration (red) or improvement (green).

This is the report's central test, and it overturns the simple version of the report's working hypothesis. If income and import dependency jointly predicted shock vulnerability, the worst outcomes would cluster in the low-income, high-import-dependency corner of this chart. They do not: the correlation between pre-shock import dependency and the change in GHI is only **r = 0.18**, and between GDP per capita and the change in GHI, **r = 0.18** as well — both far too weak, on a sample of this size, to support a claim that either variable meaningfully predicts shock vulnerability on its own.

The countries with the worst deterioration Sudan, Afghanistan, Haiti share active conflict or acute state fragility, a dimension neither axis captures. Yemen is the sharpest illustration of why a pure income/trade model misleads: despite extreme import dependency, it shows almost no further deterioration in this window, most plausibly because its hunger indicators were already at a severely elevated baseline before the shock, leaving little further room to worsen the same ceiling effect visible in the country-level inflation data. **Conflict and governance capacity, not income or trade exposure in isolation, are the dominant drivers of vulnerability to this shock in the data.**

4. Limitations

- The panel covers 30 countries over 10 years; relationships reported as correlations should be read as associations within this sample, not as universal or causal laws.
- General inflation and food inflation are near-collinear (r = 0.99) and are treated as a single signal rather than independent evidence throughout this report.

- The pre-shock "baseline" period (2015–2021) itself includes the COVID-19 disruption years; this may understate the true pre-crisis steady state and compress the measured size of the 2022–23 deterioration for some countries.
- Conflict and state fragility are identified here as the likely dominant factor by exclusion (neither income nor import dependency explains the worst outcomes) rather than through a direct conflict-exposure variable; a fragility/conflict index should be incorporated directly in the next iteration to confirm this.
- Exchange-rate and sovereign-debt exposure — a plausible amplifier for countries such as Turkey and Sri Lanka during this period — is not yet included as a variable and is a natural next addition alongside a conflict index.

5. Recommendations

- **Re-anchor targeting criteria around fragility, not just income or trade exposure.** Humanitarian and resilience funding models built primarily on income tier or import dependency will systematically misallocate toward countries that are trade-exposed but stable and under-allocate to conflict-affected states where the same shock does far more damage.
- **Build a composite early-warning score.** Combine food import dependency, income level, and a conflict/fragility index (e.g., existing fragile-states indices) into a single pre-shock vulnerability score and validate it against this shock before the next one arrives.
- **Treat stunting as a structural, multi-year indicator, not a shock-response metric.** Since single-year food inflation does not significantly predict stunting, monitoring and intervention design for stunting should track multi-year deprivation trends rather than react to a single inflationary year.
- **Add exchange-rate and debt-distress exposure to the model.** Countries undergoing currency depreciation or sovereign debt distress concurrent with a global shock face compounding pressure that current variables do not capture.

6. Conclusion

Global food price shocks do not transmit evenly, and the reason is not simply that poorer or more import-dependent countries are more exposed. It is that **conflict and governance capacity determine whether a country can absorb a global shock at all** income and trade exposure shape the baseline severity of hunger, but they are weak predictors of who gets hurt most when a new shock hits. Any resilience strategy, aid allocation model, or early-warning system built on income and trade data alone will miss the country most likely to be pushed into crisis by the next shock.